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The growing concentration of national influence in global science

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Charles J. Gomez

Scientific influence, or the capacity of ideas and concepts to shape future research, is crucial to developing and disseminating knowledge and sustained innovation. Here, using nearly 240 million scientific works published between 1990 and 2023 from OpenAlex to construct international networks of influence, discursive influence, which represents what global scientific communities consider important and worthy of investigation, is shown to be disproportionately and increasingly concentrated in a small group of resource-wealthy countries, including the United States, Canada, Western Europe and East Asia, in comparison to attributional influence. This concentration raises issues of equity and innovation in global scientific discourse, perhaps narrowing research perspectives, exacerbating biases and creating echo chambers that are associated with stifling innovation and marginalizing contributions from countries that are peripheral to global scientific discourse. The findings underscore the need to promote diverse and inclusive global research enterprises.

Influence in science occurs when ideas and concepts shape the direction, focus and evolution of future research and thinking. Such influence encompasses the ways in which scientific knowledge is created, disseminated, validated and built upon in scientific communities. It manifests in the publication of groundbreaking research that reflects the development of new theories or methodologies and even brings forth new paradigms. However, growing concern suggests that new scientific influence is disproportionately and increasingly emerging from a few resource-rich countries[1,2], and thus researchers from only a handful of places are disproportionately influencing global scientific discourse[3–6]. This might merely reflect science as a successful meritocracy—researchers worldwide sifting through new ideas and concepts, rewarding only the most promising and best ideas. Such a concentration might reflect a reward system of attention and recognition across fields. However, science is structurally plagued by the Matthew Effect[7], wherein early gains of attention, resources and recognition yield disproportionate rewards, decoupled from merit.

This is concerning for two reasons. First, sustained scientific innovation might be in jeopardy. Research into collective problem-solving warns that even if ideas and collaborators are of greater quality, monopolized attention on only select ideas or collaborators might be associated with inhibiting the diverse perspectives and ideas needed for serendipity and novel combinations of ideas[1,8,9], which might impact sustained innovation across fields. Second, there is concern regarding ensuring equity and inclusion of peripheral perspectives in science. Scientists from traditionally underrepresented, if not excluded, countries are increasingly participating in global scientific conversations. Such scientists are more likely to internationally coauthor research and publish in internationally recognized journals[10–14], and yet the reach of their ideas might be far more limited.

Influence is embodied and measured empirically using scientific output, or metadata that associate with scientific publications, the medium of discourse and the evolution of the pool of epistemic knowledge in science. Influence in science is also often conceptualized and understood as a singular measure, and yet little attention is paid to how influence manifests in these scientific outputs, particularly internationally. Influence is deployed predominately internationally in bibliometric analyses in two ways: (1) influence as discourse and (2) influence as attribution. Discursive influence[15] represents how ideas and concepts that are embodied by scientific publication terms or phrases diffuse across the scientific community's worldwide discourse by being incorporated in papers. Text data are the ideal medium to embody scientific knowledge's abstract and complex nature, where it can be extracted and measured at scale[1,15–20], making discursive influence a highly useful

School of Sociology, The University of Arizona, Tucson, AZ, USA. e-mail: cjgomez@arizona.edu

and attractive way to gauge influence. When text is extracted from abstracts or titles of papers, discursive influence distinctly reflects the focus of influencing papers and influenced papers. By examining discursive influence, ideas and concepts that are gaining attention and shaping the direction of scientific enquiry can be identified. It provides insights into the intellectual currents in the scientific community, and it highlights trends and shifts in research priorities. In contrast, attributional influence represents the recognition of work in international scientific communities through citations that reflect the networks and paths of established knowledge and those who have established it. Citations as attributional influence reflect intellectual acknowledgement of an idea's impact on the field or its author who developed the idea[21–25]. Attributional influence reveals paths of established knowledge, demonstrating how ideas are built on and which researchers or works are foundational. Attributional influence helps identify and understand the intellectual lineage of scientific ideas, as well as identify contributors to the advancement of knowledge.

Discursive and attributional influences are complementary; they demonstrate how scientific ideas not only proliferate, but gain acknowledgement, highlighting the intellectual lineage of ideas and their diffusion across global scientific discourse. Attributional influence, through citations, reflects the breadth of ideas incorporated from various sources for diverse purposes, and discursive influence represents focused research priorities. Despite its ability to shape academic discourse, the latter receives less attention, raising concerns about its influence on research trends and priorities. Parsing and understanding influence as either discursive or attributional reveals divergences between the two at the global level, with consequences not only for the growing concentration of international influence and inequality, but for sustained innovation of the scientific enterprise itself.

Using nearly 240 million papers from a public dataset of scientific publication metadata called OpenAlex[26], thousands of potential scientific concepts, ideas and phrases, and the papers that mention them in either their abstracts or titles, from 1990 to 2023, are identified. All works that cite these papers (that is, attributional influence) and those that not only cite them but also mention the same terms in their abstracts or titles (that is, discursive influence) are identified. International networks of attributional and discursive influence are constructed, finding that they correlate greatly and supporting that both are complementary.

A growing body of research suggests that scientific influence is increasingly concentrated in a small number of 'core' countries[1,6,27]. These core countries—including the United States, Canada, Western European nations and parts of East Asia, such as China—are typically resource-rich nations that house the world's top universities and produce the highest volume of scientific research. China's inclusion as a core country reflects its considerable rise over the past few decades to become a major contributor to science and technology[28–30]. In contrast, 'periphery' countries are those with fewer resources for scientific infrastructure and are often underrepresented in global scientific discourse. While researchers from the periphery are increasingly participating in international science, their work may be overlooked. This dynamic creates a stratified global scientific community where a country's position in the core or periphery can impact the visibility and influence of its research.

Where influence originates reveals a growing global imbalance. Specific paths of 'relative influence' between attributional and discursive influence reveal not only a growing disconnect between the two but also international concentration within them. The results show that attributional and discursive influence increasingly come from core countries. In these locations are the best universities that publish the most research, in comparison to those that lie on the periphery. These countries are increasingly and disproportionally concentrating influence among themselves, deciding what is important and worthy of research in the global scientific community, rather than bridging the growing divide between core and periphery countries. With growing concentration of discursive 'relative' to attributional influence, core countries might disproportionately become trendsetters in scientific research, deciding the direction of future enquiries and innovations. Core countries might also increasingly shape the global research agenda by promoting specific terms, concepts and methodologies, leading to a focus on issues more relevant to them and potentially neglecting problems pertinent to less influential areas. This might result in a homogenized scientific landscape, in which diverse research interests and perspectives are underrepresented. Such growing concentration of influence has implications not only for equity and inclusion but also for sustained novelty and breakthroughs in science[31]. Discovery is predicated on a diversity of ideas and perspectives and serendipity. Growing global conformity in scientific research from a few countries might constrain sustained innovation and threaten the scientific enterprise. In contrast, with growing and greater concentration of attributional 'relative' to discursive influence, periphery countries might see their research cited widely among themselves, not necessarily shaping leading scientific discourse. This might lead to local research priorities and contexts being overlooked in favour of topics deemed important and not integrated into current and emerging scientific conversations, limiting its immediate impact on the direction of new research.

Results

Constructing discursive and attributional influence

Much attention is paid to the concentration of attributional influence and its worrying trends[2], but the concentration of discursive influence is overlooked, even though it suggests a more concerning trend. Attributional influence is defined using citations in a paper, which serve various purposes. They acknowledge extant work, support a paper's arguments and provide context. This type of influence is characterized by wide distribution across numerous sources and reasons for inclusion[32,33]. Citations are scattered throughout a document, integrated for various reasons, and they cater to disparate objectives, making attributional influence dispersed and multifaceted. In contrast, discursive influence pertains to using terms and concepts in the abstracts or titles of academic papers that reflect their foci. Although they might appear in the body of a paper, terms in an abstract or title signify a paper's research focus, encapsulating the essence of a study's central theme. Unlike attributional influence, which amalgamates ideas from existing works, discursive influence assesses the present study's main concerns and objectives. This focal specificity makes discursive influence more troubling when overlooked, since it shapes the research agenda and the direction of scholarly discourse directly.

When countries are more discursively influential 'relative' to their attributional influence, it suggests that their research agendas, priorities and frameworks are shaping global scientific discourses disproportionally. This raises several concerns. First, it results in a narrowing of perspectives. If a few countries dominate discursive influence in academic literature, it can lead to a homogenization of research priorities. Such dominance means that unique challenges, perspectives and innovations from less discursively influential countries are underrepresented or overlooked. It also limits the diversity of ideas and approaches that are crucial to addressing complex global issues. Second, it exacerbates biases in research priorities. Discursive influence shapes what is considered important and worthy of investigation in academic communities. If some countries exert greater discursive influence, there is a risk that research agendas become biased towards their interests, concerns and viewpoints. Such bias affects funding allocation, publication opportunities and the overall direction of scientific enquiry, potentially ignoring issues that are critical to other regions of the world. A concentration of discursive influence also leads to the creation of echo chambers, in which ideas, methodologies and theories are reinforced and alternative viewpoints are marginalized. Such

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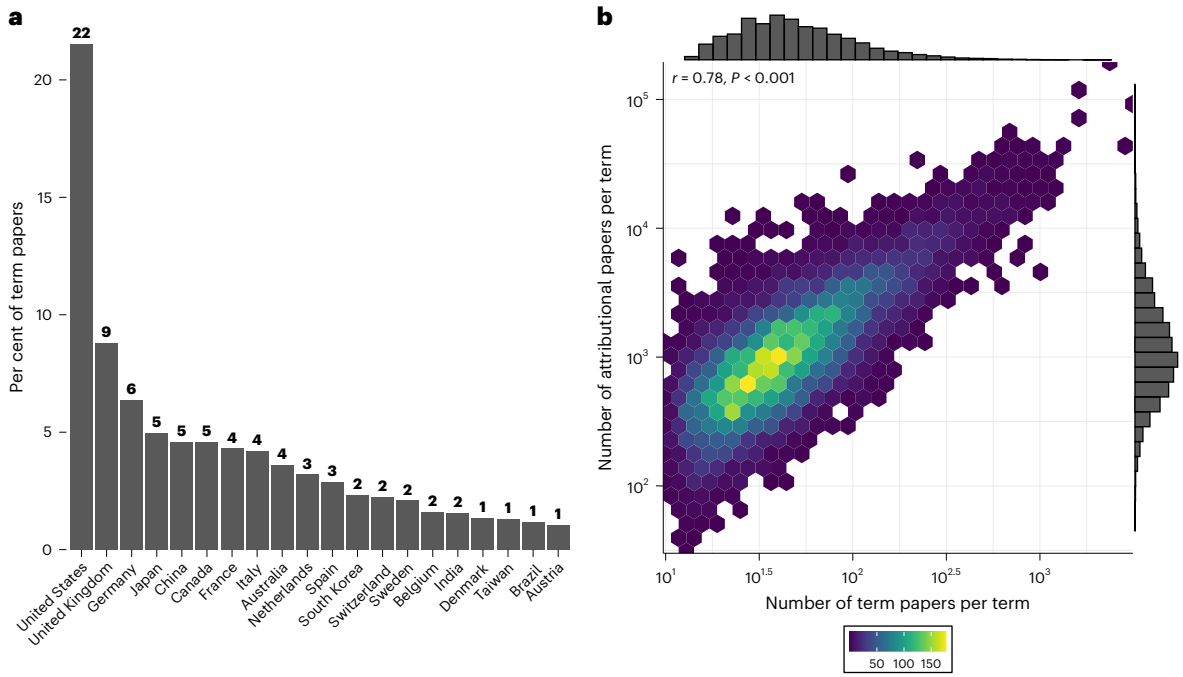


Fig. 1 | The distribution of term papers. a, Per cent of term papers (*y* axis) from the top 20 countries that authored the most 'term papers' (*x* axis). **b**, Number of 'term papers' per term (*x* axis) and the number of attributional papers per term, or the number of citations of term papers within 10 years of it being published (*y* axis). Both the *x* and *y* axes are plotted on a $\log_{10}$ scale. Data points are terms visualized as a hexagonal heat map, where lighter colours reflect higher frequency. Histograms are plotted on the margins. Statistical test (**b**): Pearson correlation (two-tailed), $r = 0.78$, $P < 0.001$, 95% CI 0.77–0.78, $n = 10{,}046$ terms.

reinforcement stifles innovation and critical thinking, leading to less dynamic and responsive academic environments. Disproportionate discursive influence from a few countries might skew policy recommendations and interventions towards those that favour influential countries' contexts, ignoring the nuances of local contexts in less influential regions. Researchers in areas peripheral to mainstream scientific communities might find it increasingly difficult to break into established discursive frameworks if they are dominated by a few. This creates barriers to entry for new ideas and researchers, especially those from underrepresented regions, making it more difficult for them to gain recognition and contribute to global scientific discourse.

Attributional influence represents the breadth of sources and ideas recognized through citations, and a higher 'relative' discursive influence signifies a concentration of power in shaping what is studied, discussed and valued in global science. Such a 'relative' imbalance poses challenges to the global research enterprise's diversity, equity and inclusiveness and its ability to address global challenges with nuanced, varied perspectives.

To this end, a majority-vote ensemble of several popular key phrase extractors written in `Python` 3.8 that isolate phrases with at least three characters, ranging from unigrams (singular terms) to five-grams (five consecutive terms) is used, from the titles and abstracts of 239,036,158 million papers published between 1990 and 2023 from OpenAlex. All papers that incorporate the term in their abstracts or titles are identified, and these papers are called 'term papers'. Only terms whose first appearance in the data is at least in the year 1990 and appears in at least 10 'term papers' in OpenAlex are considered. 'Term papers' must have been cited to be included in the analysis. In the Discussion and 'Limitations and further research' sections, the advantages of this approach over considering only the first paper in the data in which a term appears are discussed, reflecting the introduction of a novel idea or concept, since this approach is used in other studies of influence[34]. English is largely the common language of scientific publications, evidenced over the past 20 years in many scientific fields[35–37], so papers published in English are considered when identifying and extracting terms. This choice is discussed in detail in the Methods, Discussion and 'Limitations and further research' sections.

For each 'term paper' *i*, all future works that cited it within a 10-year window after publication are identified. A limited timeframe is used because the influence of a specific paper is most direct within a certain period after its publication. Over longer periods, researchers may use terms from older papers without citing them, as such terms have become common knowledge in the field. A 10-year window was chosen as a conservative measure to capture the peak of a paper's influence while minimizing the noise from concepts that have become broadly integrated into the scientific lexicon. Citing papers that use the same term in their abstracts or titles as the cited 'term paper' *i* are called 'discursive papers'. Works that merely cite 'term paper' *i* but do not have the term in their abstracts or titles are called 'attributional papers', reflecting how attributional influence through citations acknowledge the ideas in cited papers and employ them in the various ways outlined. In Supplementary Note 1, how 'attributional' and 'discursive papers' are identified to construct attributional and discursive influence networks, respectively, is further outlined.

This distinction is important. A paper merely having the same term in its abstract or title is not a clear indication of discursive influence, but when a paper cites the 'term paper' that has the same term in its abstract or title (within the 10-year time window), it is a strong indication of discursive influence. 'Discursive papers' and 'attributional papers' are mutually exclusive; a citing paper that shares the same term as the 'term paper' is counted as a 'discursive paper', not an 'attributional' one. 'Discursive papers' themselves are also 'term papers' that associate with their own set of 'attributional papers' citing it, and their own set of 'discursive papers' citing it and with the shared term in its abstract or title. This process was repeated for all subsequent 'term papers' for all terms, resulting in 10,046 terms that appear in over half a million (567,258) 'term papers' that originated from 213 countries.

Figure 1 reports descriptive statistics and trends of the 'term papers' used to construct international networks of influence. Figure 1a plots the per cent of 'term papers' on the *y* axis from the top 20 countries

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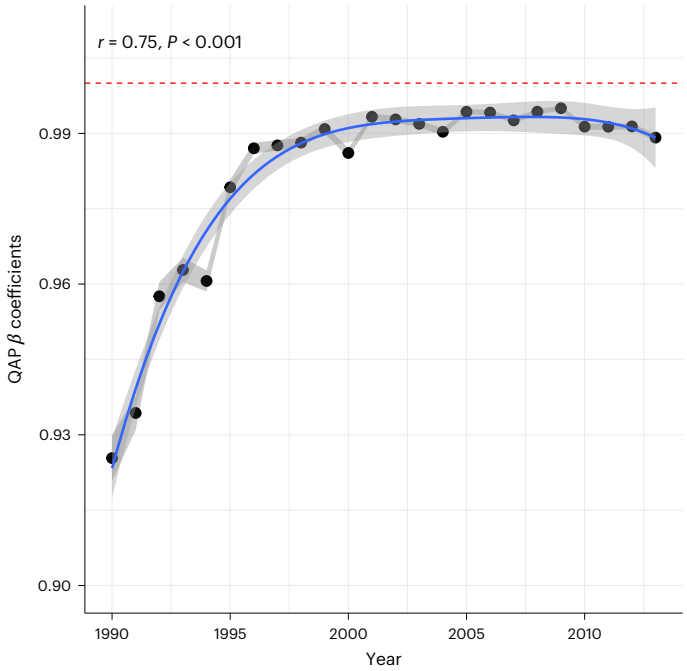


Fig. 2 | QAP model of attributional influence and discursive influence over time. β coefficients from the QAP regression models of international networks of discursive influence regressed onto international networks of attributional influence plotted by year from 1990 to 2013. On the *y* axis are the β coefficients (interpreted as *z*-scores), plotted as data points, with the shaded region as the standard errors for each year on the *x* axis. Data points represent yearly QAP regression β coefficients (mean of bootstrapped distribution); shaded region shows ±1 s.e. The red dashed line indicates parity ($\beta = 1$). Statistical test: QAP regression with 10,000 permutations, two-tailed; all yearly coefficients significant at $P < 0.001$. Correlation between β and year: $r = 0.75$, $P < 0.001$, 95% CI 0.49–0.88. Mean β across years = 0.98 (95% CI 0.97–0.99).

that authored the most 'term papers' on the *x* axis, ranked left to right from highest to lowest percentage. These countries represent over 80% of all 'term papers' used during analysis that met various criteria (for example, a term that appeared in at least 10 papers, a term that first appeared after 1990 and so on.). Figure 1b plots the distribution of the number of 'term papers' (that is, the papers with a particular term in its abstract or title) on the *x* axis, with the number of 'attributional papers' per term (that is, the number of citations these 'term papers' received within 10 years of their publication) on the *y* axis. Both axes use a logarithmic scale ($\log_{10}$) to represent the range of values. Terms are represented as data points but visualized using a hexagonal heat map, in which lighter colours indicate higher frequencies, showing where clusters of terms, as data points in the plot, are densest (Pearson correlation coefficient of $r = 0.78$, $P < 0.001$, 95% CI 0.77–0.78). Histograms independently display the distribution of terms along both axes. Multiple examples and details of terms are discussed in Supplementary Note 2. The average number of 'term papers' per term is 67, meaning that the average number of papers with the same term is 67. The average number of 'attributional papers' per term (the number of citing 'attributional papers' that 'term papers' received within the 10-year window for a given term, averaged across terms) is 1,933 papers. Thus, for each term, the number of 'attributional papers' citing its 'term papers' is 1,933.

Quadratic assignment procedure models of international networks of influence

The analysis shows that international networks of attributional and discursive influence are highly correlated. A linear regression model for networks, called quadratic assignment procedure (QAP), was used to calculate this correlation. Yearly international networks of attributional and discursive influence from 1990 through 2013 were constructed. This is shown in greater detail in Supplementary Note 3. The analysis was stopped at 2013 since a 10-year window was used, and the version of the data that were used from OpenAlex ends in 2023. The international networks for attributional influence were constructed using 'term papers' and 'attributional papers', or papers that cite the 'term paper' but do not have the same term in the abstract or title. They were constructed as the number of 'attributional papers' that cite a 'term paper' published within the 10-year, post-publication time window. The countries that are associated with the 'term paper' and those that are associated with the 'attributional paper' are noted. Each carries a weight of '1'. In cases of internationally co-authored papers, the weighted edge of '1' is split among the countries proportionately. These directed weighted international edges are combined and summed across all 'term papers' and their associated 'attributional papers' each year, creating the international network of attributional influence. In Supplementary Note 4, the edge weights are normalized for both attributional influence and discursive influence by terms, where the edge weights are percentages per term. Then, the percentage edge weights are summed by country dyads in each year. The results are the same.

Figure 2 plots β coefficients from 24 QAP models (that is, one model per year), starting in 1990 and continuing to 2013. The dependent variable is the international network of discursive influence for year *t*, and the independent variable is its corresponding international network of attributional influence for year *t*. To render the two networks comparable across time and within a year, the edge weights are converted to *z*-scores (that is, mean-centre and network edge weights normalized by standard deviations), which converts the QAP intercept coefficient to null, allowing comparisons across models. Each model produced a β coefficient that reflects how a one-standard-deviation increase in attributional influence corresponds to a change in discursive influence, measured in standard deviations. The QAP model also produces standard errors, shown by the shaded area, that are associated with each β coefficient, which were statistically significant at a two-tailed 0.05 significance level.

The models demonstrate that discursive and attributional influences associate highly with one another as international networks; countries that influence attributionally also influence discursively in equal measure. A value of 1 reflects parity; one standard deviation in attributional influence corresponds to one standard deviation in discursive influence, shown in Fig. 2 as the horizontal dashed red line that intercepts the *y* axis at 1. With a Pearson correlation coefficient of $r = 0.75$ ($P < 0.001$, 95% CI 0.49–0.88), the average value of these β coefficients across time is 0.98 (95% CI 0.97–0.99), meaning that a one-standard-deviation increase in the weighted edges of an attributional influence international network associates, on average, with a one-standard-deviation increase in the weighted edges of its corresponding discursive influence international network. This is noteworthy because international discursive influence networks were constructed from a mutually exclusive set of papers (that is, 'discursive papers') from international attributional influence networks (that is, 'attributional papers'). This high correlation suggests an apparent parity at the aggregate global level; a country's attributional influence seems to move in lockstep with its discursive influence. However, this macro-level convergence presents a puzzle, as it masks growing divergences when influence is examined between specific groups of countries, a dynamic explored in the following sections.

'Where' influence originates reveals a growing global imbalance. To explain this contradiction, the 'type' of country is used in both international networks of attributional and discursive influence. Figure 3 plots the centralities of countries in the international networks of discursive and attributional influence. For attributional influence, a country's centrality score in the network is its degree centrality, or the number of works that cite a paper with a term within the 10-year

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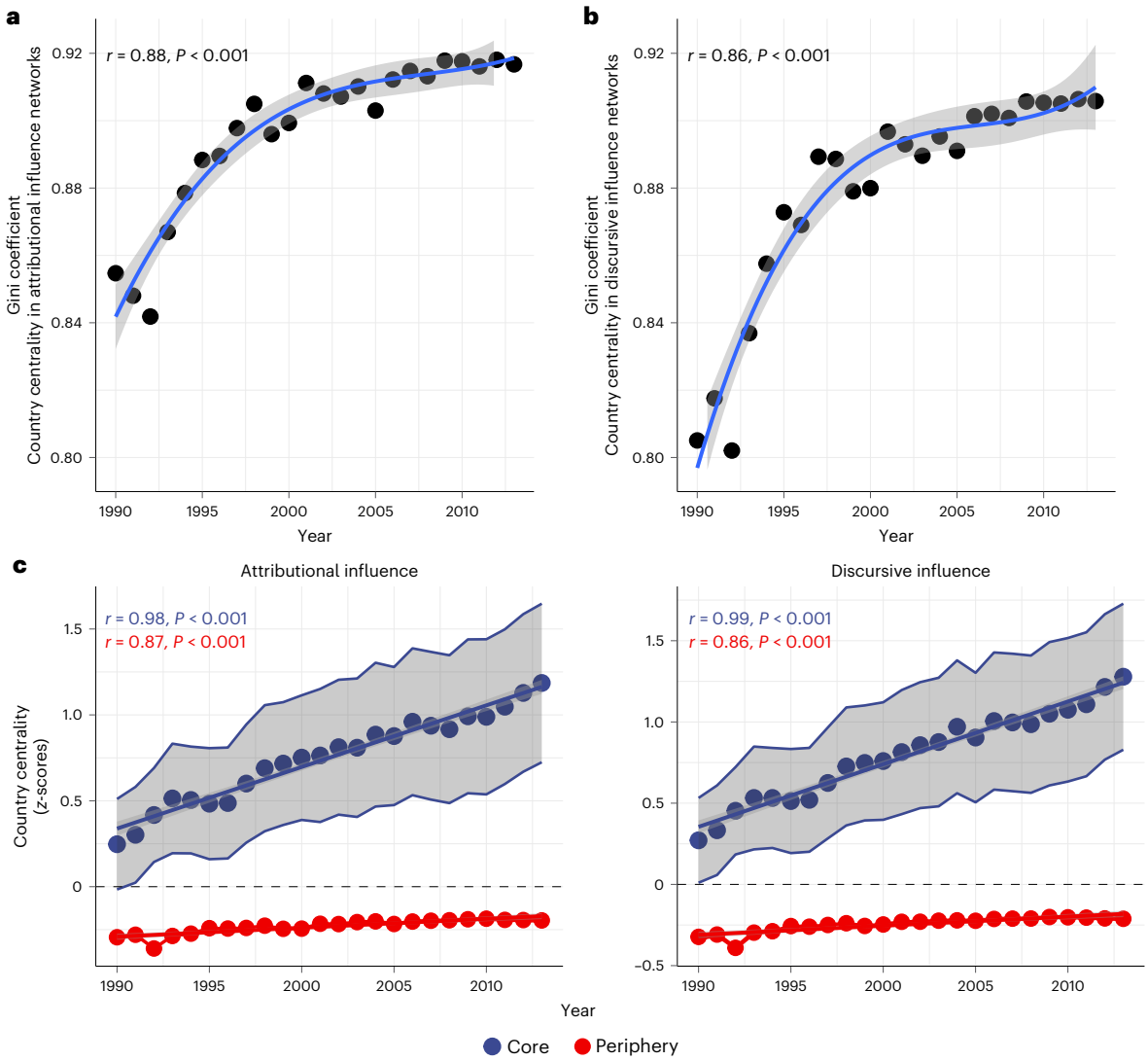


Fig. 3 | Centralities of countries and their concentration in international networks of attributional influence and discursive influence plotted by year from 1990 to 2013. a,**b**, Gini coefficient (*y* axis) among country centralities in both international networks of attributional influence (**a**) and discursive influence (**b**) over time (*x* axis). **c**, Average country centralities (*y* axis, as *z*-scores) for both core and periphery countries in both networks over time (*x* axis). Statistical test: Pearson correlation (two-tailed) with $n = 24$ yearly observations. In **a**: $r = 0.88$, $P < 0.001$, 95% CI 0.74–0.95. In **b**: $r = 0.86$, $P < 0.001$, 95% CI 0.70–0.94. In **c**, data points represent yearly mean centrality values (mean of country centralities for each group); shaded regions show ±1 s.e.m. For core countries: attributional $r = 0.98$, $P < 0.001$, 95% CI 0.95–0.99; discursive $r = 0.99$, $P < 0.001$, 95% CI 0.95–0.99. For periphery countries: attributional $r = 0.87$, $P < 0.001$, 95% CI 0.74–0.95; discursive $r = 0.86$, $P < 0.001$, 95% CI 0.72–0.94.

time window after it was published and scaled yearly as a *z*-score. For discursive influence, a country's centrality score in the network is its degree, or the number of works that cite a paper with a term within the 10-year time window after it was published that has the same term in its abstract or title, also scaled yearly as a *z*-score.

The Gini coefficient[38], calculated across all centralities for each year, is a standard measure of inequality. A value of 1 reflects full inequality and 0 reflects perfect equality in an economy. The Gini coefficient is used to reflect whether influence is increasingly and disproportionately concentrated among a handful of countries. Since the coefficients require all values to be positive, the minimum centrality value is added to all centrality values for a year so that no value is negative.

In Fig. 3a,b, Gini coefficients applied to the centralities for attributional and discursive influence are rising over time. Each data point is a Gini coefficient of the distribution of centralities. Figure 3a shows the coefficients applied to attributional influence network centralities, and Fig. 3b shows discursive influence network centralities. Both show that the concentration of influence increasingly originates from some countries in both networks over time (attributional influence with a Pearson correlation coefficient of $r = 0.88$, $P < 0.001$, 95% CI 0.74–0.95, and discursive influence with a Pearson correlation coefficient of $r = 0.86$, $P < 0.001$, 95% CI 0.70–0.94). Figure 3c shows the type of country in which concentration is most prominent. It plots the average centrality for core and periphery countries for both attributional and discursive influence networks over time. Each data point is the average centrality for core or periphery countries each year. Core countries appear in blue, and periphery countries are in red. The left pane plots attributional influence and the right pane plots discursive influence. Core countries are increasingly central in both networks at a much higher rate than periphery countries (for core countries: attributional influence with a Pearson correlation coefficient of $r = 0.98$, $P < 0.001$, 95% CI 0.95–0.99; and discursive influence with a Pearson correlation coefficient of $r = 0.98$, $P < 0.001$, 95% CI 0.95–0.99; for periphery countries: attributional influence with a Pearson correlation coefficient of $r = 0.88$, $P < 0.001$, 95% CI 0.74–0.95; and discursive influence with a

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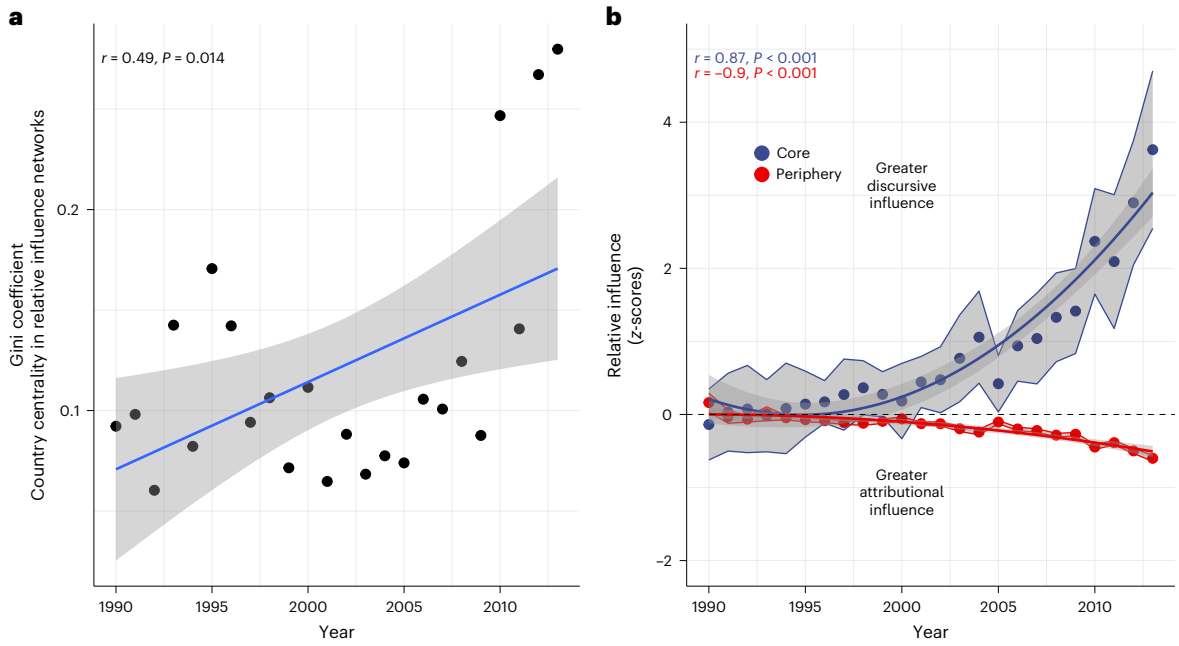


Fig. 4 | Centralities of countries and their concentration in international networks of 'relative influence' plotted by year from 1990 to 2013. a, Gini coefficient (*y* axis) among country centralities in international networks of 'relative influence' over time (*x* axis). **b**, Average country centrality (*y* axis, as *z*-scores) in the 'relative influence' networks for core and periphery countries over time (*x* axis). Data points reflect average centrality values for core and periphery countries, and the shaded regions are standard errors (±1 s.e.m.). Statistical test: Pearson correlation (two-tailed) with $n = 24$ yearly observations. In **a**, $r = 0.49$, $P = 0.014$, 95% CI 0.11–0.75. In **b**, core countries: $r = 0.87$, $P < 0.001$, 95% CI 0.74–0.95; periphery countries: $r = -0.90$, $P < 0.001$, 95% CI −0.96 to −0.77.

Pearson correlation coefficient of $r = 0.87$, $P < 0.001$, 95% CI 0.72–0.94). Both international networks of influence are increasingly unequal and concentrated in a few core countries.

International networks of relative influence

To resolve the contradiction between the high global correlation of influence types and the observed concentration of influence in core countries, it is necessary to isolate their meaningful divergences. To show this, 'relative influence' is defined as the difference between every dyad (that is, country-to-country) in the international networks of attributional influence and its corresponding dyad in the international network of discursive influence. The work on citational lensing[1], which identifies growing deviations between what is being researched between countries and how infrequently they are being cited, is used. For year t, the dyad in the discursive network ($\text{Discursive}^t_{i\to j}$), where country i is the national affiliation of the paper with the same term in the abstract or title as the citing paper affiliated with country j, is matched with the corresponding dyad in the attributional network ($\text{Attributional}^t_{i\to j}$), where papers affiliated with country i receive citations from papers affiliated with country j. The weighted edge values of these dyads are subtracted across all dyads in these networks for year t to construct a 'relative influence' network, defined as:

$$\text{Relative}^t_{i\to j} = \text{Discursive}^t_{i\to j} - \text{Attributional}^t_{i\to j} \quad (1)$$

A positive value of $\text{Relative}^t_{i\to j}$, where $\text{Discursive}^t_{i\to j} > \text{Attributional}^t_{i\to j}$, indicates that country i exerts more discursive influence on country j 'relative' to its attributional influence on country j. Conversely, a negative value of $\text{Relative}^t_{i\to j}$, where $\text{Discursive}^t_{i\to j} < \text{Attributional}^t_{i\to j}$ indicates that country i exerts more attributional influence on country j 'relative' to its discursive influence on country j. However, individual dyadic values between countries i and j are not as revealing as the centralities of countries i and j in an international network of 'relative influence' constructed from $\text{Relative}^t_{i\to j}$ dyads for each year. Centrality scores in 'relative influence' international networks reveal which countries exert more discursive or attributional influence overall and whether these trends diverge over time.

Figure 4 is similar to Fig. 3 but instead shows plots for international networks of 'relative influence'. Centralities are defined as the degree of centrality of each country and region in the 'relative influence' networks. Figure 4a plots the Gini coefficient of centrality scores in the international network of 'relative influence' each year, where the *y* axis is the Gini coefficient and the *x* axis is time. Rising Gini coefficients indicate that countries are not only increasingly exerting more of one influence over another, but that some countries are also disproportionately doing so (Pearson correlation coefficient of $r = 0.49$, $P = 0.014$, 95% CI 0.11–0.75). Figure 4b plots the two distributions of average centrality scores from the international 'relative influence' network as data points (core countries in blue and periphery countries in red). Values above the *x* axis reflect countries with greater discursive influence 'relative' to attributional influence, and values below the *x* axis reflect countries with greater attributional influence 'relative' to discursive influence. The shaded region is the associated standard error for each trend. Core countries are, on average, above the *x* axis, moving away exponentially, meaning they are increasingly far more discursively influential 'relative' to their being attributionally influential (core country centralities with a Pearson correlation coefficient of $r = 0.88$, $P < 0.001$, 95% CI 0.74–0.95). Periphery countries are, on average, just below the *x* axis, exerting slightly more attributional influence 'relative' to discursive influence, but to a far lesser degree than core countries' growing proclivity towards discursive influence (periphery country centralities with a Pearson correlation coefficient of $r = -0.90$, $P < 0.001$, 95% CI −0.96 to −0.77). Over time, core and periphery countries diverge in their forms of influence, and the overall imbalance between the two types of influence grows.

Core and periphery trends in international relative influence networks

The origins and targets of international influence are examined here to address two contradictory trends—growing convergence between

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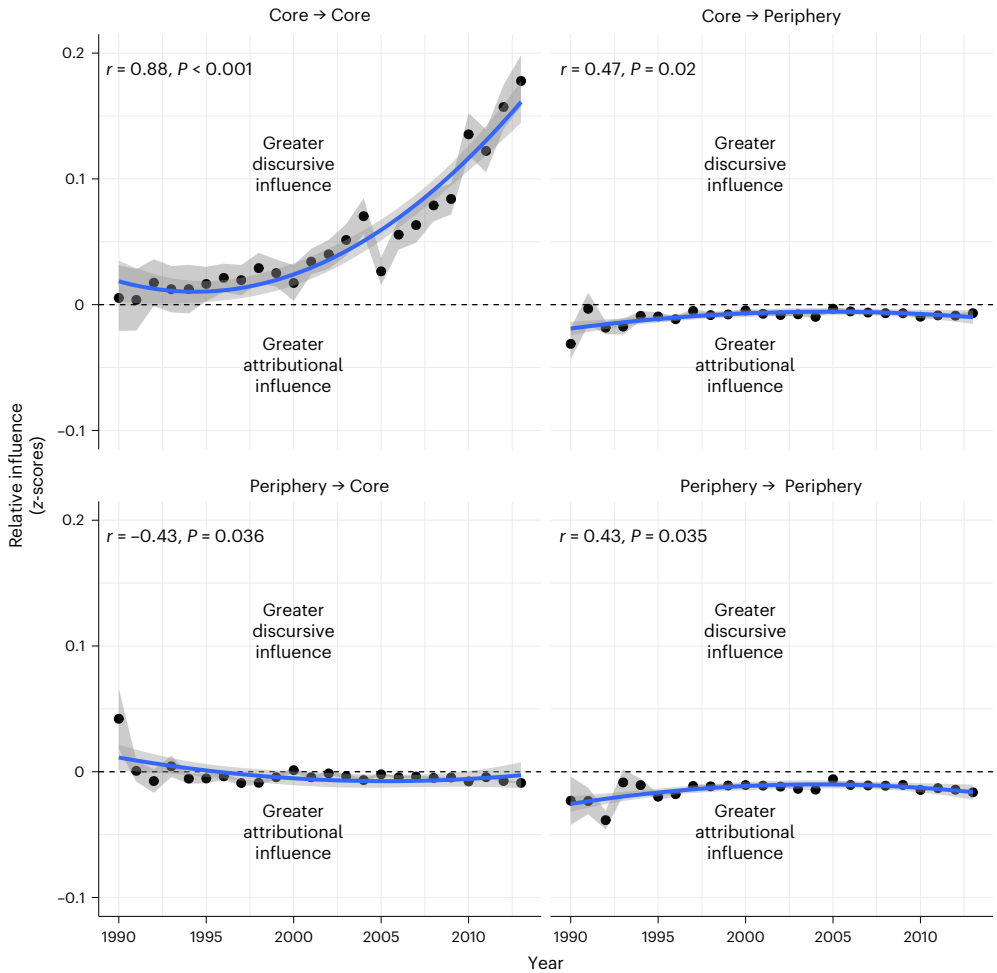


Fig. 5 | 'Relative influence' across and between core and periphery regions. Average 'relative influence' (*y* axis, as *z*-scores) for core and periphery countries in one of four panes (starting at the top left and moving clockwise): Core → Core, Core → Periphery, Periphery → Periphery, and Periphery → Core, over time (*x* axis). Shaded regions are standard errors (±1 s.e.m.). Data points represent yearly mean relative influence values (mean of country-pair dyads for each group). Statistical test: Pearson correlation (two-tailed) with $n = 24$ yearly observations. Core → Core: $r = 0.88$, $P < 0.001$, 95% CI 0.74–0.95. Core → Periphery: $r = 0.47$, $P = 0.020$, 95% CI 0.08–0.73. Periphery → Core: $r = -0.43$, $P = 0.036$, 95% CI −0.71 to −0.03. Periphery → Periphery: $r = 0.43$, $P = 0.035$, 95% CI 0.04–0.71.

attributional and discursive influence globally, but growing divergence between types of country (that is, core and periphery). Figure 5 shows the average of international dyads of 'relative influence' between and among core and periphery countries over time in one of four panes (starting at the top left and moving clockwise): Core → Core, Core → Periphery, Periphery → Periphery, and Periphery → Core. The *y* axis plots the 'relative influence' over time on the *x* axis. 'Relative influence' is measured as *z*-scores, including standard errors as the shaded region. Core countries exert far more discursive influence among themselves (Core → Core with a Pearson correlation coefficient of $r = 0.88$, $P < 0.001$, 95% CI 0.74–0.95). The 'relative influence' among periphery countries and between core and periphery countries, reflected with dyads Periphery → Periphery, Core → Periphery, and Periphery → Core, are mostly close to the *x* axis, where discursive and attributional influences are mostly near parity (that is, $y = 0$).

The core-country trends are shown in Fig. 6a, which visualizes and organizes vertical network arcs between 10 regions and countries. Figure 6b unpacks and re-visualizes Fig. 6a as a heat map. Most definitions, including sociology and international relations and political economy work[1,39,40], typically define 3 regions that encompass most of the world's geopolitical and economic activities: North America, Western Europe and East Asia. Building from this and the United Nations Regional Groups classification scheme, a 10-category classification scheme was created. The first set of regional categories classifies core countries as part of 1 of 3 regions—core countries in the Commonwealth (that is, Canada, Australia, New Zealand and the United Kingdom), core countries in Europe (that is, countries in Western Europe, that is, Germany, Austria, Belgium, Denmark, Finland, France, Iceland, Ireland, Italy, Netherlands, Norway, Portugal, Spain, Sweden and Switzerland, including Israel) and core countries in Asia (that is, Japan, South Korea, Taiwan and Singapore). The remaining countries in the periphery are defined as Periphery Europe, Periphery Asia, Latin America, Africa and Oceania. The United States and China are included as their own nodes to reflect their distinct and unique status in global science[29,30,41,42]. The full list of countries by region appears in Supplementary Note 5.

Figure 6a plots the average 'relative influence' arcs between these 10 regions for 2001 (left) and 2013 (right). 2001 was selected because it represents the mid-point of the data (that is, 1990 to 2013) and is one of the last years in which the 'relative influence' centralities of core and periphery countries were mostly the same. Each set of arcs is divided into a left hemisphere, which are 'relative influence' arcs between regions that are negative, reflecting greater attributional influence, and a right hemisphere, which are 'relative influence' arcs between regions that are positive, reflecting greater discursive influence. This is similar to Figs. 4 and 5. The arcs are dark red on the left, reflecting greater attributional influence (that is, negative 'relative influence') and dark green on the right, reflecting greater discursive influence (that is, positive 'relative influence'). Thicker arcs reflect

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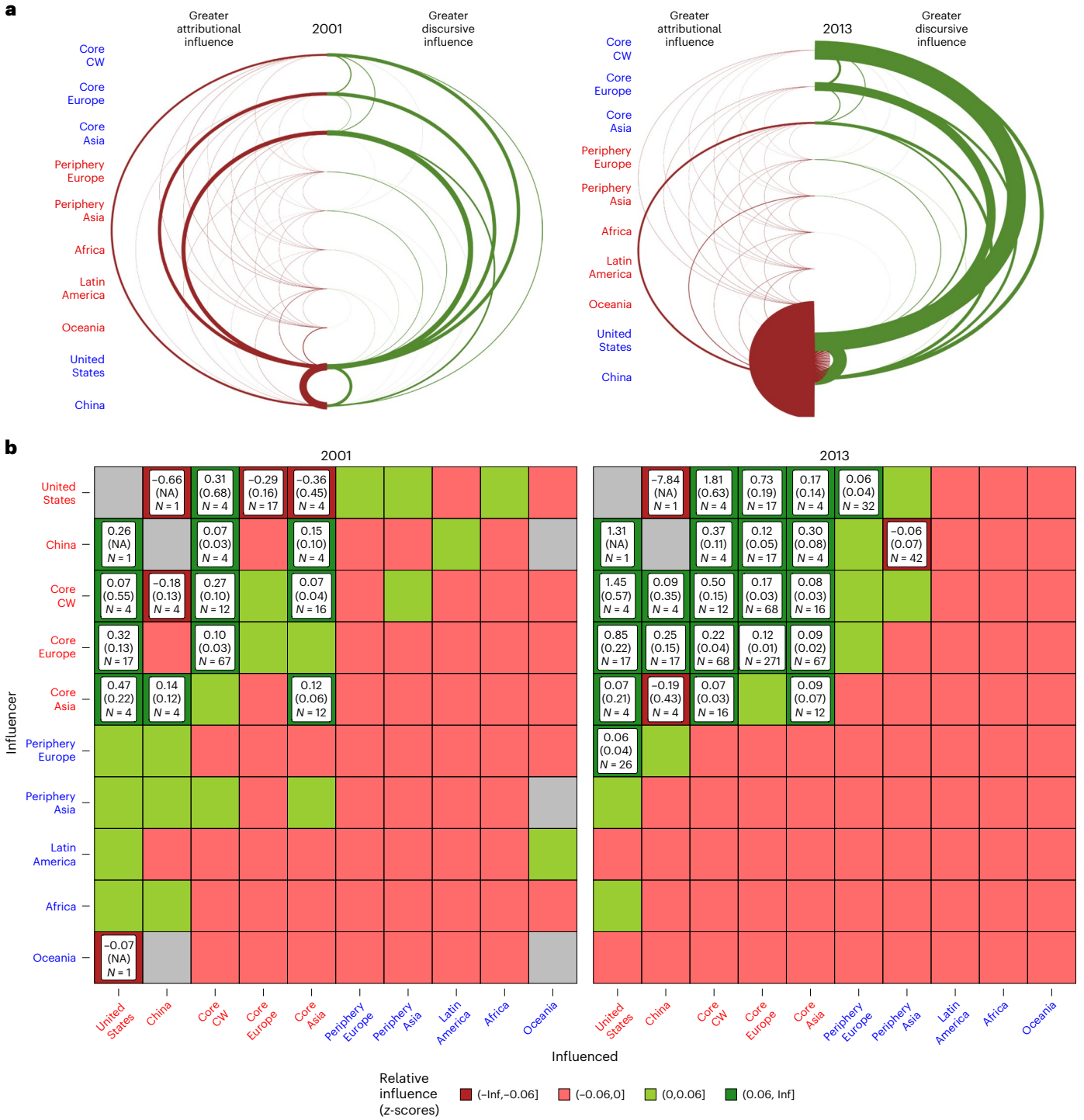


Fig. 6 | 'Relative influence' across and between regions. a, Average 'relative influence' arcs among 10 regions and countries for 2001 (left) and 2013 (right). Core regions and countries are in blue, and periphery regions are in red. Arcs are divided into the left hemisphere, which includes 'relative influence' arcs between regions that are negative, reflecting greater attributional influence, and the right hemisphere, which includes 'relative influence' arcs between regions that are positive, reflecting greater discursive influence. Arcs are dark red on the left, reflecting greater attributional influence, and dark green on the right, reflecting greater discursive influence. Thicker arcs reflect greater values in 'relative influence', either negative (greater attributional influence in dark red in the left hemisphere) or positive (greater discursive influence in dark green in the right hemisphere). CW stands for 'Commonwealth Countries'. **b**, The values in **a** but as a heat map, highlighting extreme 'relative influence', with text boxes containing standard errors in parentheses and the number of dyad observations, *N*. Extreme values of 'relative' attributional influence are dark red, and extreme values of 'relative' discursive influence are dark green. Grey tiles represent missing values. The influencer region is on the *y* axis, and the influenced region is on the *x* axis.

greater 'relative influence', either for more attributional influence on the left hemisphere in dark red or more discursive influence on the right hemisphere in dark green. Core regions and countries are blue, and periphery regions are red.

Corroborating results shown in Fig. 5, the dark green arcs that reflect greater discursive influence grew across core regions and countries between 2001 and 2013. These large, dark green arcs on the right originate almost entirely in, and are directed towards, core regions

and countries in blue. The periphery regions in red are absent among these international arcs of 'relative influence'. The United States–China dyad is an exception, since the large dark red arcs that reflect greater attributional influence on the left grew between 2001 and 2013.

Mimicry and the displacement of national research agendas

To test whether the concentration of influence comes at the cost of displacing local research agendas, the extent to which a country's scientific focus begins to mimic that of a more influential country is measured. First, a 'national research agenda' for each country in each year is defined. This agenda is represented as a term-frequency vector derived from the *n*-grams present in the titles and abstracts of all papers published by authors from that country in a given year. For internationally co-authored papers, the term counts are fractionally allocated across the authors' countries, consistent with the method for measuring influence described earlier.

Using these yearly national research agendas, a 'mimicry ratio' was devised to measure the displacement of a country's scientific focus. This ratio compares how similar a country's current research agenda is to an influential country's past agenda, relative to how similar it is to its 'own' past agenda. The similarity between agendas is calculated using cosine similarity. To capture the 'past', a 5-year moving average of a country's research agendas was used. The resulting ratio is defined as:

$$\text{Mimicry}_{i\to j}(t) = \frac{\langle \cos(V_j(t),\ V_i(t-n))\rangle_{n\in(1,2,3,4,5)}}{\langle \cos(V_j(t),\ V_j(t-n))\rangle_{n\in(1,2,3,4,5)}} \quad (2)$$

$\text{Mimicry}_{i\to j}(t)$ represents the mimicry score for an influenced country (j) relative to an influential country (i) at a specific year (t). This score is a ratio calculated using a cosine similarity (cos), which measures the cosine similarity between two vectors. The numerator compares the term-frequency vector of country j in the current year, $V_j(t)$, to the term-frequency vector of the influential country i over the 5-year period preceding the current year, $V_i(t-n)$, where $n \in (1, 2, 3, 4, 5)$, reflecting country i's agendas from $t-5$ to $t-1$. The denominator then compares country j's current agenda, $V_j(t)$, to its own term-frequency vector from the same previous 5-year period, $V_j(t-n)$. The numerator represents the average similarity between the influenced country's current research agenda (country j at year t) and the influential country's past agenda. The denominator represents the average similarity of the influenced country's current agenda to its own past agenda. A value greater than 1 indicates that country j has become more similar to country i's past research agenda than to its own, suggesting that its local scientific focus is being displaced. Conversely, a value less than 1 indicates that country j is maintaining its own distinct research trajectory, remaining more similar to its own past work than to that of the influential country.

Figure 7 illustrates the trends in research agenda mimicry, measured by cosine similarity, between and among core and periphery countries from 1995 to 2013 in one of four panes (starting at the top left and moving clockwise): Core mimicking core in the past (Core → Core), Periphery mimicking core in the past (Core → Periphery), Core mimicking periphery in the past (Periphery → Core) and Periphery mimicking periphery in the past (Periphery → Periphery). The y axis plots the average mimicry dyad between the different types of country over time on the x axis. A strong and statistically significant positive trend is evident in the Periphery mimicking core in the past panel (a Pearson correlation coefficient of $r = 0.66$, $P = 0.002$, 95% CI 0.30–0.86), suggesting that the research agendas of periphery countries are increasingly mimicking those of core countries. Core countries' research agendas have become increasingly similar to one another over time (Core mimicking core in the past, a Pearson correlation coefficient of $r = 0.96$, $P < 0.001$, 95% CI 0.90–0.98). In contrast, there is a significant negative trend for influence originating from the periphery; core countries are becoming less similar to periphery countries (Core mimicking periphery in the past, a Pearson correlation coefficient of $r = -0.62$, $P = 0.004$, 95% CI −0.84 to −0.23), and periphery countries are also diverging from one another (Periphery mimicking periphery in the past, a Pearson correlation coefficient of $r = -0.69$, $P = 0.001$, 95% CI −0.87 to −0.34). This implies that local research agendas of periphery countries are being displaced by core countries.

While these mimicry patterns suggest a hierarchical flow of influence, a more direct test of agenda displacement involves examining the lifecycle of scientific concepts based on their origin. An analysis tracking the global prevalence of terms originating from core versus periphery nations reveals a stark divergence in their trajectories (Supplementary Note 9). Terms introduced by core countries overwhelmingly achieve wider and sustained adoption over time, while terms originating in the periphery stagnate and fail to diffuse into the global discourse. This finding provides direct evidence that the observed convergence is asymmetrical, wherein core-country research agendas not only expand but actively displace or overshadow those from the periphery.

Discussion

Modern science has always been, to some extent, a global enterprise[43–47]. However, the promise of globalization at the end of the twentieth century was that science and its ideas would be unencumbered by borders[43,46,48,49], and it would be more inclusive of perspectives from researchers in countries often relegated to the periphery. This vision manifested in increased international collaborations, mobility and greater participation, but the assumption was that it would also foster more equal discourses.

This study conversely highlights a significant trend in the concentration of a type of influence that disproportionately shapes the direction of future enquiries and innovations in global scientific communities. A select group of countries that comprises the core of global science dominates discursive influence, a phenomenon in which a small number of countries disproportionately shapes scientific ideas and frameworks, and has critical implications for the future of scientific innovation and equity. When influence is understood in global science not as a single measure but in two distinct forms—discursive and attributional influence—a growing 'relative' imbalance emerges.

Discursive influence, which reflects the focus and integration of concepts, ideas and frameworks in global scientific discourse, and attributional influence, marked by the recognition of scientific ideas used to build new ones, both show that core countries, such as the United States, Western Europe and some East Asian nations, are centralizing their influence. Discursive influence reflects what is introduced into the scientific episteme and embodies a deeper, more indirect layer of influence than attributional influence. Citation accumulation is often driven by Matthew Effects, which benefit elite researchers and universities[1,7,50]. Reference 2 found that scientists in high-ranking universities in high-income core countries such as the United States, Western Europe and East Asia increasingly resource themselves, leading to disproportionate exertion of discursive influence on countries relegated to the periphery of global science. Given the nature of discursive influence, its impact on what is researched in global science, and the fact that discursive influence is a more profound marker of scientific influence that is far outpacing attributional influence, these trends are worrying.

Observed centralization is evident in the growing measures of concentration (Figs. 3–7), and yet more troubling is the growing 'relative' disproportionate concentration of influence on discursive influence exclusively among a few core countries (Figs. 4–6). Core countries appear to be becoming insular, increasingly influencing each other and directing trends in science, while periphery countries are excluded from central scientific discourse. To explain this, 'relative influence' is

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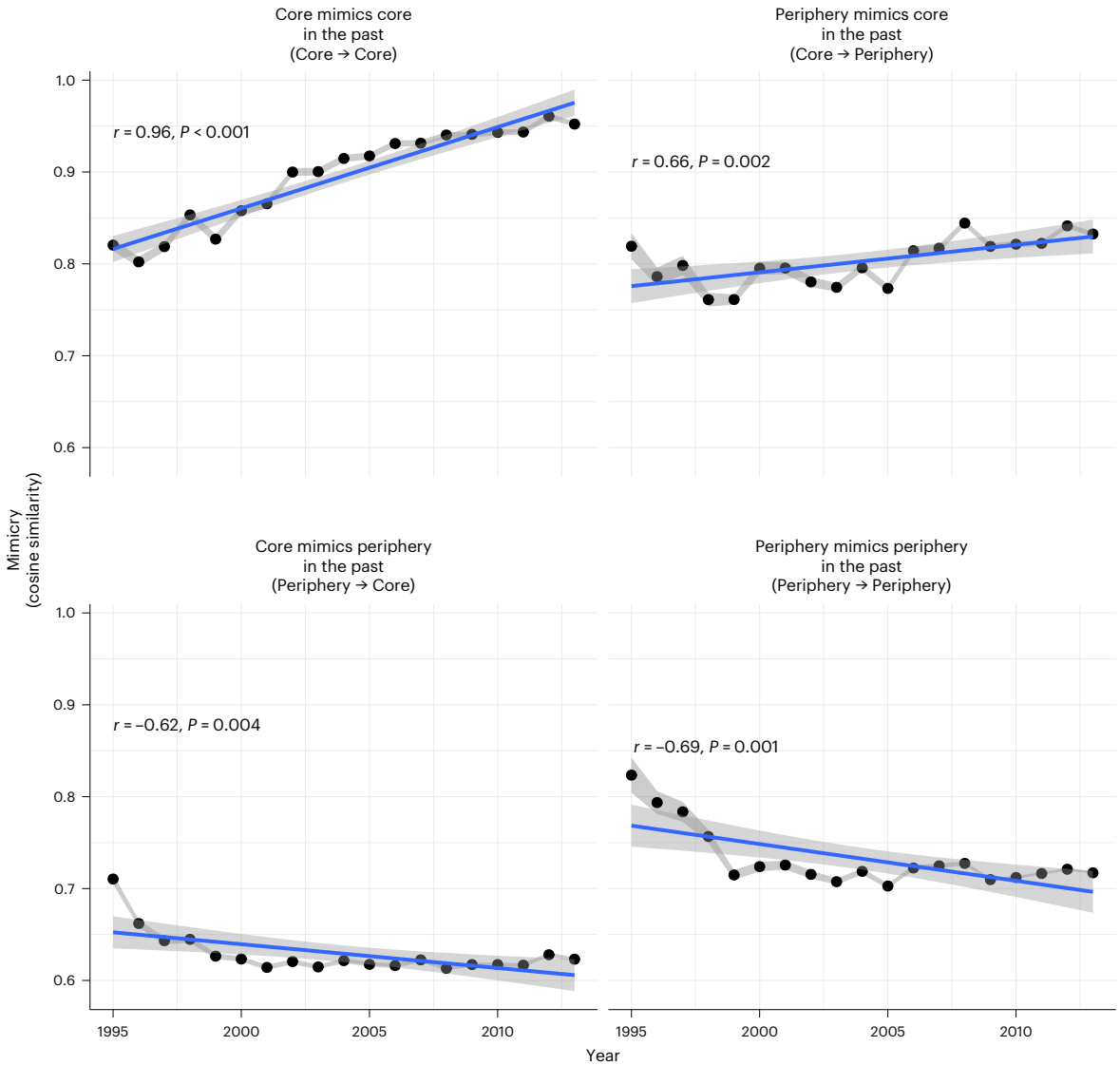


Fig. 7 | Mimicry as the displacement of national research agendas. Mimicry between and among core and periphery countries in one of four panes (starting at the top left and moving clockwise): Core mimicking core in the past (Core → Core), Periphery mimicking core in the past (Core → Periphery), Core mimicking periphery in the past (Periphery → Core) and Periphery mimicking periphery in the past (Periphery → Periphery). The *y* axis plots the average mimicry dyad between the different types of country over time on the *x* axis. Solid blue lines are the lines of best fit. Shaded regions are standard errors. Data points represent yearly mean mimicry values (mean of country-pair dyads for each group); shaded regions show ±1 s.e.m. Statistical test: Pearson correlation (two-tailed) with $n = 19$ yearly observations (1995–2013). Core → Core: $r = 0.96$, $P < 0.001$, 95% CI 0.90–0.98. Core → Periphery: $r = 0.66$, $P = 0.002$, 95% CI 0.30–0.86. Periphery → Core: $r = -0.62$, $P = 0.004$, 95% CI −0.84 to −0.23. Periphery → Periphery: $r = -0.69$, $P = 0.001$, 95% CI −0.87 to −0.34.

defined as the difference between the corresponding nodes and ties in international networks of attributional influence and its counterpart network of discursive influence. Variance in the distribution of country centralities increase in the 'relative influence' network, as a way to measure inequality, shows a dramatic increase as core and periphery countries diverge in their forms of influence over time; core countries are more discursively influential in comparison to their being attributionally influential, and core countries are exerting more discursive influence only among themselves over time. This growing divide and concentration might lead to a situation in which scientific research priorities and innovations are decided on the basis of the interests and perspectives of a few countries, neglecting global challenges that require diverse approaches and solutions. A slight increase is evident in the growing 'relative' attributional influence from periphery countries (Fig. 4), which might reflect greater acknowledgement of these countries. However, core countries dominate both discursive and attributional influences (Fig. 3).

This growing divide is further evidenced by an analysis of research agenda mimicry, which measures the similarity of scientific topics discussed by different groups of countries (Fig. 7). The results show a strong positive trend of core countries becoming increasingly similar to each other over time, effectively creating a more homogeneous and insular scientific mainstream. Furthermore, this core agenda appears to be displacing those of the periphery: The research of periphery countries is increasingly mimicking the past research of core countries. This influence is unidirectional, as core countries are simultaneously becoming 'less' similar to the research produced in the periphery. This suggests that the concentration of discursive influence is not just a matter of recognition but an active process of agenda-setting that pulls periphery science into the orbit of core countries, while core countries' own discourse becomes ever more consolidated.

This centralization of influence raises a critical question: Does it reflect epistemic dominance and structural dependency, or is it the result of researchers from the periphery voluntarily engaging with

globally relevant topics? While the distinction is nuanced, the findings reveal a one-way flow of influence: Periphery countries increasingly mimic the research agendas of the core, but core countries do not reciprocate, instead diverging from periphery research topics. This non-reciprocal pattern is more consistent with a hierarchical structure of dependency than with a symmetrical, voluntary global dialogue. Moreover, the fact that core countries' research agendas are also becoming increasingly similar to one another suggests the formation of a consolidated and insular mainstream that defines what are considered 'globally relevant' topics, thereby constraining the choices available to researchers on the periphery.

Nevertheless, this concentration may simply reflect the natural development of 'global' research frontiers led by core countries, rather than an exclusionary structure that displaces local research agendas. This then raises another critical question: Does this concentration of influence from core countries erase all agency from the periphery, or can periphery countries still exhibit regional leadership in specific fields? Could the observed pattern simply reflect a natural convergence on 'global' research frontiers, rather than an exclusionary structure that displaces local research agendas?

While greater capacity in core countries is an undeniable factor, the findings of this study suggest that the observed patterns cannot be explained by resource levels alone. The divergence between relative discursive and attributional influence already indicates that core countries' role extends beyond high-volume production to disproportionate agenda-setting. The analysis of research mimicry provides stronger evidence for the presence of an exclusionary structure (Fig. 7), where the unidirectional flow of influence and mimicry is particularly telling. The research agendas of periphery countries do not appear to influence the core's agenda. This demonstrates a form of epistemic displacement, where the core frames the scientific narrative, displacing the periphery's own agenda, supporting the conclusion that the observed patterns are a structural phenomenon with implications for epistemic equity and diversity.

An analysis separating 'global' terms from two types of 'local' term was also performed: (1) those used exclusively within periphery countries and (2) those that reference topics salient to or about periphery countries. (For the full analysis and figures, see Supplementary Note 6.) The findings reveal that while the periphery maintains a small, resilient internal discourse on its own exclusive topics, the core's disproportionate, trend-setting influence dominates the wider discourse, including their own. Thus, the findings do not deny the existence of regional or local interests but suggest that these pockets of focus struggle to scale up and shape the broader, global scientific conversation; instead, the periphery's agenda is dominated by the core's agenda. Taken together, this suggests that the global scientific structure actively filters or devalues knowledge originating from the periphery and instead supplants it with the core's research agenda (Fig. 7), which has profound and negative implications for equity and stifles the diversity of ideas needed for sustained scientific innovation.

The findings underscore the implications of the concentration discussed above. First, the potential for sustained innovation might be jeopardized. Innovation thrives on diverse perspectives and the serendipitous combination of ideas from various sources. Science that is dominated by a few countries might limit diversity of ideas, leading to a homogenized research agenda that stifles novel discoveries. The Matthew Effect, in which initial advantages in resources and recognition disproportionately benefit established players, exacerbates this issue, creating a feedback loop that intensifies the concentration of influence. Second, the exclusion of periphery countries from core scientific discourse raises concerns about equity and inclusiveness. Researchers from these regions increasingly participate in international collaborations and publish in prominent journals, but their ideas might not gain the same attention or recognition as those from core countries. Such disparity would marginalize important research perspectives and limit the global applicability of scientific advancements.

Global scientific research is growing at an extraordinary rate, and thus the success of modern science, as an engine of technological breakthroughs and innovation, is attributed to it being a global enterprise in which discoveries are disseminated widely in journals and, increasingly, at international conferences in scope and locale. As discoveries are published, other researchers build on extant findings, leading to breakthroughs elsewhere. Flows of information between researchers across countries drive novel ideas and innovation. This work underscores and extends the concerns of a growing imbalance in influence in global science[2,5,51–53], so much so that influence might be even much more skewed than is already suggested by the diffusion and citation networks that demonstrate bias towards prominent countries that lie at the core of global science. Reference 54 shows that prestige strongly influences how ideas diffuse and drive inequality, since countries with elite scientists and universities increasingly command attention and focus. This is concerning because many underrepresented groups produce higher rates of scientific novelty, while their contributions are devalued and discounted[9]. Exposure to diversity leads to better outcomes in collective problem-solving, such as scientific research[8].

Many studies claim that churning of scientific breakthroughs and innovations is declining[31,55], which might result from the sheer complexity and the volume of scientific knowledge and research. However, a few countries' increasingly overt influence might be contributing to these trends. Sustained breakthroughs in science are predicated on novelty and diversity[9]. New combinations of knowledge that would have otherwise been lost can lead to untapped and undiscovered lines of enquiry. Most exploration and exploitation research underscores that searching for new and better solutions during collective problem-solving is achieved through exploration rather than exploitation[8,56]. If, however, discursive influence in global science is increasingly and disproportionately exerted by a few researchers in very prominent countries, exploitation becomes the mechanism through which discourse occurs. The consequences might be a world in which global scientific agendas across fields are based on the work of these highly influential sources. The quality of work produced by these sources is probably quite high, and yet multiple potential lines of enquiry that would have otherwise been pursued and recombined in innovative ways by researchers in other countries are removed from the discourse. The search for new lines of enquiry is thus pulled from a limited pool of concepts, ideas and approaches, the result of which might be growing conformity of research foci and agendas due to this imbalanced influence. Future research should test this, but the implications are clear and dire. If the concentrated influence of a few countries continues at its current pace, sustained scientific breakthroughs and innovations might be untenable and threaten the vibrancy of the global scientific enterprise.

To address some of the issues that favour some regions over others, policymakers and institutions could encourage researchers to engage directly with relevant and impactful research from diverse geographic regions, which might mitigate the concentration and divergence of influence. International funding agencies and evaluative bodies could reassess criteria for measuring research impact and quality, considering the diversity of contributions from these regions. Academic and science policymakers could also encourage collaborative research that involves institutions and researchers from diverse geographical locations to broaden these regions' knowledge pools, although the impacts of such interventions remain unknown. The current study contributes to ongoing dialogues on the dynamics of influence and its implications for academia. These tentative suggestions represent a starting point for future research.

Limitations and future research

Several limitations in this study are evident in the operationalization, construction and analysis of influence, providing opportunities for future research. Several major areas of concern are outlined, explaining why they are unlikely to change conclusions drastically.

Term significance. Some terms and phrases might hold more importance in some knowledge domains versus others. This importance to a field is not directly accounted for, but only the volume of use across countries. Future research should incorporate a field-level weighting scheme for terms and phrases. Combining the importance of a term in a field with the importance of terms and phrases would provide a more nuanced model of diffusion. However, results are unlikely to change too much, since the volume of 'term papers' is a good proxy for importance in the field.

Different interpretations of relative influence. The 'relative influence' used in this study can be interpreted in multiple ways, potentially leading to disparate conclusions about the nature and direction of scientific influence. For example, greater discursive influence could be interpreted as being under-recognized by another country, as a country receives fewer citations in proportion to the degree to which it discursively influences the direction of scientific conversations. Alternatively, greater attributional influence could be interpreted as a country being over-recognized 'relative' to how much it discursively influences science, as it receives more citations than it should. However, this is unlikely; core countries are by far more discursively influential 'relative' to their being attributionally influential.

Core countries dominate in terms of both discursive and attributional influence, indicating that they set scientific agendas and receive recognition for contributions. Such dual dominance suggests that these countries are not only shaping the direction of scientific research through influential concepts and ideas, but are also being acknowledged and cited across the global scientific community. Observed patterns of influence thus probably reflect the true prominence and impact of core countries in the global scientific discourse, rather than any form of under- or over-recognition. This reinforces the argument that scientific influence is highly concentrated among a few countries, which has implications for the inclusivity and diversity of global scientific research. The approach used in this study, distinguishing discursive and attributional influence, provides a clearer and more nuanced understanding of how scientific ideas are propagated and recognized internationally.

Citation window selection. The choice of a 10-year citation window to measure influence is a methodological decision intended to balance capturing a concept's diffusion with the eventual assimilation of terms into common knowledge. While using various citation timeframes after publication is common in bibliometric studies[1], the dynamics of influence can vary across different fields. A shorter window might under-capture the long-term impact of foundational ideas, while a longer or unlimited window could conflate direct influence with the general usage of established terms. Future work could explore how varying window lengths might reveal different facets of scientific influence across diverse disciplines.

Unit of analysis. This study uses country as the primary unit of analysis to investigate the international and geopolitical structure of scientific influence. This approach allows for an examination of how national scientific systems interact on a global scale. However, countries are heterogeneous, and internal variance exists, particularly within 'core' nations that house highly stratified academic systems. The influence attributed to a country such as the United States or the United Kingdom is probably driven disproportionately by a small number of elite institutions (for example, Ivy League or Russell Group universities). Given the high correlation found between discursive and attributional influence networks, it is highly likely that the same elite universities that rank highest in traditional citation impact are also the primary drivers of the discursive influence that are identified. The concentration of both forms of influence within these institutions is a manifestation of the Matthew Effect, where prestige and recognition beget further rewards.

A valuable extension of this work would be to disaggregate the analysis to the university or institutional level. Such an approach would enable a more nuanced understanding of how institutional prestige and resources contribute to the concentration of discursive influence. A key question for future research is whether the same high-prestige universities that excel in traditional metrics, such as field-weighted citation impact, also lead in discursive influence, or whether there are notable exceptions. Identifying institutions with, for instance, high citation impact but lower relative discursive influence (or vice versa), could help to further elaborate and specify the mechanisms of scientific agenda-setting, as the initial analysis of the United States–China dyad suggests.

Drivers of mimicry. An important limitation and direction for future work is to more fully distinguish between two potential drivers of research mimicry (Fig. 7): the genuine adoption of core-country topics versus the strategic necessity for researchers in periphery countries to align their work with the thematic interests of international journals to be published. The observed trend could reflect a strategic shift by researchers in periphery countries who align their work with topics prominent in core countries to increase their chances of being published in prestigious international journals. Alternatively, it may represent a more fundamental adoption of core country research agendas across a nation's entire scientific enterprise, influencing even research intended for domestic publication. The OpenAlex dataset, similar to many others, is more likely to index English-language and internationally focused journals, which makes it difficult to disentangle these two possibilities. Future research could investigate this distinction by comparing the publication patterns in global datasets with those in regional and national repositories. In addition, qualitative studies with researchers in periphery countries could provide direct evidence on the motivations shaping their research agendas.

Scientific production capacity. This study does not include a direct control for the vast differences in scientific production capacity between nations. It is possible that the observed concentration of influence is, in part, a reflection of the sheer volume of research produced by core countries. However, the study's primary metrics were designed to mitigate this effect and uncover patterns that are not reducible to production volume alone. The 'relative influence' metric, for instance, focuses on the difference between discursive and attributional influence, revealing the character of a nation's influence rather than its raw volume. Furthermore, the mimicry analysis is longitudinal and self-referential, comparing a country's research agenda to its own past trajectory, thereby isolating shifts in focus from raw increases in output. While these methodological choices were meant to ensure that the findings are not simply an artefact of production volume, future research could build on this work by incorporating explicit controls, such as normalizing influence by a country's total scientific output or number of researchers. This would enable a more precise determination of how influence operates independently of production capacity.

Data biases. Any analysis at this scale is subject to limitations in the underlying data. OpenAlex, while highly valuable, contains inconsistencies inherent to large, dynamically aggregated databases, such as missing institutional affiliations, incomplete or erroneous author data and incomplete citation records for certain periods or journals. These issues represent known challenges in computational social science and can introduce noise into the analysis.

Beyond these record-level inconsistencies, another issue is a potential bias towards English-language journal outlets and the representation of non-Western citation indices. OpenAlex inherits publication data from the Microsoft Academic Graph (MAG), which probably disproportionately indexes English-language and Western outlets. The result might be a biased sample of global science, in which authors

worldwide seek to publish in journals that are influenced disproportionately by Western gatekeepers and reviewers. However, strong anecdotal evidence suggests that an unbiased dataset, even if possible, would yield a similar result.

Nonetheless, OpenAlex is among the most representative of the major repositories. It is also unique in that it is updated every 2 weeks on the basis of web scrapes. This not only includes the latest publications, but retroactively backfills past publications, offering an advantage over Scopus and Web of Science regarding how OpenAlex updates its repository dynamically and directly from the Internet, where many excluded, non-Western and non-English venues probably reside. However, ref. 57 argue that perhaps tens of thousands of non-predatory journals are not indexed in these repositories. MAG, the predecessor to OpenAlex, is as representative as other major repositories such as Scopus and Web of Science[58–60]. Although these data might be biased, the data are among the best available, and the results presented here are unlikely to lead to radically different conclusions.

The non-Western and non-English journals and publications included in OpenAlex and used during analyses might still be representative samples of these excluded data. 'Relative influence' would thus only be further corroborated with these excluded non-Western and non-English journal and publication data. If researchers from core countries disproportionately discursively influence others who publish there, then including any omitted outlets would corroborate the trends discussed here, since these omitted outlets would enhance the disparities in 'relative influence' between discursive and attributional influence shown in Fig. 3.

English is considered the common language for modern scientific publications[35–37], although national non-English journals exist in many fields. Nonetheless, English is a distinct and consistent feature of international science, especially since many governments are pushing for stronger international presence among their researchers and universities. For example, most of the top 20 countries associated with 'term papers' listed in Fig. 1 are non-English-speaking countries, including Brazil, India, Germany, China, South Korea and Japan, and only 7% of abstracts in the data are non-English.

Country affiliations. An author's affiliated country on large, multi-author papers may not accurately represent international influence, as it overlooks the complexity of co-authorship networks that can distort the results. To address the potential distortion caused by the complexity of co-authorship networks, a robustness check was conducted by restricting the analysis to domestically collaborated papers only. Specifically, whether restricting all papers—'term papers', 'discursive papers' and 'attributional papers'—to domestic collaborators (that is, papers without international collaborations) when creating international networks of attributional and discursive influence changes the findings was tested. This analysis is discussed in Supplementary Note 7. The main conclusions, however, remain unchanged, suggesting that the observed patterns are not merely an artefact of international collaborations.

In addition, not every author contributed to ideas presented in a paper, especially regarding large collaborations, in which there are highly demarcated divisions of academic labour. Nevertheless, future research should use the recent incorporation of metadata of author attribution schemes. For example, the Contributor Roles Taxonomy (CRediT) system, which several publishers use, offers such an opportunity (see ref. 61 for a complete history of CRediT). By instead focusing on the country of affiliation of authors who developed a paper's central ideas, as opposed to methodologists, those who conducted the experiments, those who secured funding and so on, more precise international influence networks can be constructed. This opens new lines of enquiry, in which new diffusion models based on methodologists or funders can decompose the trends here on the basis of academic divisions of labour.

Another concern derives from authors with multiple international affiliations. For example, an author from country A who is visiting a laboratory in country B and who maintains both affiliations equally on papers seemingly affiliates with both countries A and B. This situation reflects a broader epistemological issue of how to account for knowledge diffusion—is the physical location where an idea was developed its origin (that is, the visited laboratory), or does primary affiliation reflect an author's deeper intellectual connection? Many journals require that the location where the work was done be listed as the author's primary affiliation, but how consistently this is enforced, if at all, is unclear. Future research should address this critical issue more thoroughly, but these instances appear to be the exception in OpenAlex metadata. For example, ref. 62 finds that with ORCID profile data, very few scientists maintain dual international affiliations.

Origins of terms. Several approaches use the first appearance of a term in scientific publications and its subsequent diffusion to measure influence. The analytic approach used here instead employs all papers in which a term appears ('term papers'), but is limited to papers cited by papers with the same term in the abstract or title ('discursive papers'). This method better reflects the real-world dynamics of how scientific ideas gain attention, prominence and influence in scientific discourse. The traditional method of focusing on the first appearance of a term as a measure of influence has several limitations. The first occurrence of a term does not necessarily correspond to its most impactful point of influence. The initial introduction of a term might go unnoticed or be underappreciated in an academic community, only gaining prominence when subsequent, more influential papers use and refine the concept. By analysing all subsequent papers that use the term, this approach captures a broader, more dynamic view of how the term permeates and influences a scientific community; it provides a more nuanced and detailed view of influence by examining the broader use of terms in subsequent publications. This method acknowledges that scientific influence is not a single event but a cumulative process, with ideas gaining attention through repeated use, citation and refinement in academic literature.

Citation of a subsequent, more influential paper that popularizes, or more accurately uses, the term is a more comprehensive indicator of discursive influence than merely focusing on the first paper in which it appeared. Subsequent use and citation of the term in other influential works indicate its integration into scientific discourse and its impact on shaping research agendas and knowledge structures. The paper that introduces a term might not accurately capture the term's true origin or its conceptual evolution. New terms often express ideas with long pre-histories, emerging from a continuum of thought and incremental advancement, rather than a single seminal work. Including subsequent papers in the construction of influence thus allows more comprehensive understanding of the term's conceptual lineage and its role in the evolution of scientific ideas. Many repositories do not capture the true first appearance of a term. Reference 34 uses the first appearance of a term in Web of Science as a measure of diffusion, including exemplar terms and their dates of origin. However, most of the terms can be identified in even earlier works. Empirically, focusing on the flow of 'discursive papers' as a measure of influence mitigates this issue by accounting for broader, more practical dissemination and adoption of ideas in scientific communities across countries. In Supplementary Note 8, the main results remain unchanged when the data are limited to just 'term papers', the papers in which the term first appears, and when the criterion of ten 'term paper' appearances is removed.

The China–United States dyad. A noted earlier, the United States–China dyad is an exception, reflecting the growth of mutually greater attributional influence. As shown in Fig. 6b, the United States exerts greater 'relative' attributional influence over China (−7.84), while China exerts greater 'relative' discursive influence over the United States (1.31). For

decades, China incentivized and prioritized publishing in international (that is, Western) outlets[63,64]. China's rise in citation accumulation is the result of it now curating its own publication ecosystem[65–67]. This dyad thus reflects China's unique position in the global science landscape. Future research should investigate the mechanisms underlying this mutual influence in greater detail, examining how policies, collaborations and publication practices contribute to increasing citation rates between these two countries. Future research should also consider the broader implications of this dyad to global scientific equity and innovation. Understanding these dynamics would provide valuable insights into how powerful emerging scientific producers, such as China, can integrate into and reshape the global scientific landscape.

This dyad serves as a prime example of how discursive and attributional influence can operate as distinct mechanisms. The finding that the United States exerts greater relative attributional influence on China while China exerts greater relative discursive influence on the United States demonstrates that agenda-setting is not merely a by-product of citation volume. It underscores that it is possible for a nation to be highly cited (attributional recognition) while simultaneously having its research agenda shaped by another (discursive dominance).

Methods

The study is based on publicly available bibliographic data from OpenAlex and did not involve interaction with or intervention on human participants. Therefore, institutional review board approval and informed consent were not required for this study. The core analytical strategy consisted of three main steps, which are detailed in the Results section. First, the processed term and citation data were used to construct annual international networks of attributional influence and discursive influence. Second, QAP regression models were used to assess the correlation between these two types of influence network over time. Finally, country-level centrality measures and Gini coefficients were calculated to analyse the concentration of influence and its evolution within the global scientific system.

Quadratic assignment procedure

Dependent and independent variables are matrices that represent networks, rather than continuous, categorical, or ordinal variables. Similar to how linear regression indicates the influence of one variable on another while controlling for covariates, QAP measures the association between a unit change in an independent variable network and a dependent variable network. Hence, QAP is an ordinary least squares regression that regresses the dependent network matrix on the independent network matrix. However, relational network data are not independent, and independence is necessary in linear regression models for estimates to be unbiased. To address this, QAP permutes the nodes in the dependent network matrix (that is, the rows and columns of the matrix) multiple times to create a bootstrapped distribution of randomized dependent network matrices. This creates multiple linear relationships between a randomized dependent matrix and independent network matrices. The regression coefficients from this bootstrapped distribution can be compared to the regression coefficient of the data to test for statistical significance under the null hypothesis that the rows, and simultaneously the columns, of the dependent network matrix are matched randomly with those of the independent network matrix.

Dataset generation and processing

The analysis is based on a static snapshot of the OpenAlex database, with all data current as of 1 December 2023. OpenAlex[26] is the next generation of one of the largest and most expansive repositories of academic publications, MAG[68,69], which contains extensive metadata on scientific publications, including citations, co-authorships, field designations and text from abstracts and titles. Using a fixed snapshot ensures that the study is fully reproducible and not subject to the database's continuous updates. The full dataset was downloaded and hosted on Amazon Web Services (AWS), where it was queried using AWS Athena to efficiently process the nearly 240 million publication records used in the study. OpenAlex curates extensive metadata from scientific publications, including abstracts, titles, authors, their institutional affiliations (including nations) and citation networks, which form the basis of this analysis. The text from titles and abstracts are used, offering several advantages over using full texts, which vary across fields. Some fields emphasize method over theory, and others have strict length criteria. Abstracts, however, summarize the most important concepts in a paper, and the number of words in abstracts is nearly consistent across fields. The texts are either originally in English or were subsequently extracted from their original language.

Phrase extraction and selection

A majority-vote ensemble of several popular key phrase extractors written in `Python` 3.8 was used. It isolates terms that are at least three characters long, ranging from unigrams (singular terms) to five-grams (five consecutive terms), from the titles and abstracts of over 240 million papers from OpenAlex. The text of each paper's abstract and title are combined, applying an ensemble of four phrase extractors commonly used in `Python`: `TextRank`, `Multi-RAKE`, `YAKE!` and `Textacy`. `TextRank` is an algorithm for text summarization and keyword extraction inspired by Google's PageRank algorithm to rank websites in search engine results. `Multi-RAKE` (Rapid Automatic Keyword Extraction) is an extension of the `RAKE` algorithm, designed for keyword extraction from text data. `RAKE` is a popular algorithm that identifies key phrases in text by analysing the frequency of word appearances and their co-occurrences with other words. It is particularly useful for extracting keywords that represent the main themes or topics in text without requiring extensive preprocessing or external resources such as corpora. The `Multi-RAKE` variant includes enhancements or adaptations to the original `RAKE` algorithm that allow it to be more versatile, efficient and capable of handling multiple texts or languages simultaneously. `YAKE!` (Yet Another Keyword Extractor) is an unsupervised, automatic keyword extraction algorithm designed to identify the most relevant words and phrases in a text document, without needing external knowledge or pre-labelled data. It differs from other keyword extraction algorithms by focusing on statistical features derived from the text itself, rather than relying on linguistic or semantic information. This approach allows `YAKE!` to be language independent and applicable to any text without preprocessing. `Textacy` is a `Python` library designed to process and analyse text data based on the `spaCy` library. `Textacy` extends `spaCy`'s features with additional functionalities for information extraction, including extracting phrases from text. The phrase extraction feature in `Textacy` identifies segments of text that hold particular importance or convey ideas more densely than individual words might.

The phrases extracted from each phrase extractor were compared, and a phrase was associated with a paper only if at least three of the four extractors also extracted the same phrase. Alternative spellings were identified (for example, missing adjectives, plural forms, Commonwealth English spelling versus US English spelling and so on). Noise was probably introduced during this approach, and bias might have been introduced into the measure of discursive influence. For example, papers that used a diffused term might reference it differently. A popular concept in sociology and social network analysis is 'strength of weak ties', but this concept might appear as 'weak tie' or 'weak ties'. The terms and phrases were alphabetized, and subsets of diffused terms or phrases were identified in other terms in the corpus. ChatGPT-4o was then used to identify potential superfluous phrases, and terms that reference the same thing were combined. For validation, a team of coders went through the finalized list of terms that met inclusion criteria (that is, the phrase needed to appear in the title or the abstract of at least ten papers, and the first appearance of a term

needed to begin appearing around 1990) to remove other phrases or combine phrases that referred to the same thing.

The language of an abstract was identified, and only English abstracts were included in the analysis. Including non-English terms would require translating them into English, and many argue that doing so might introduce new biases[70], so they were not included in the analysis. However, this omission is unlikely to change overall results, given the small size of non-English abstracts.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All code and metadata constructed for this project from OpenAlex are available on the Harvard Dataverse at https://doi.org/10.7910/DVN/ZSZRKK (ref. 71). The repository includes a detailed README file that describes the data processing and analysis pipeline to ensure full reproducibility.

Code availability

All analysis code is available in the Harvard Dataverse repository at https://doi.org/10.7910/DVN/ZSZRKK (ref. 71). The repository includes Python scripts for data processing, R scripts for statistical analysis and visualization, and documentation describing the complete analysis pipeline.

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Author contributions

C.J.G. collected the data, analysed them, created the figures and wrote the paper and Supplementary Information.

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Competing interests

The author declares no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Charles J. Gomez.

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Corresponding author(s): Charles J. Gomez

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data were queried using Amazon Web Services (AWS) Athena from a snapshot of the OpenAlex db.
Data analysis	Analysis was performed using Python 3.8. Code is available on the Harvard Dataverse.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

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All data and metadata constructed for this project from OpenAlex are available on the Harvard Dataverse at https://doi.org/10.7910/DVN/ZSZRKK. The repository includes a detailed README file that describes the data processing and analysis pipeline to ensure full reproducibility.

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Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used.* *Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected.* *Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Reporting on race, ethnicity, or other socially relevant groupings	*Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status).* *Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.)* *Please provide details about how you controlled for confounding variables in your analyses.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Bibliometric/network analysis of attributional & discursive influence in global science.
Research sample	Metadata for nearly 240 million scientific works (1990 and 2023) from the OpenAlex db.
Sampling strategy	Complete static snapshot of the OpenAlex db as of December 1, 2023.
Data collection	Collected by downloading a snapshot of OpenAlex and hosting it on AWS and running queries.
Timing	Analysis covers 1990 to 2023. The data snapshot is current as of December 1, 2023.
Data exclusions	English abstracts/titles; terms in at least ten "term papers"; term appeared starting 1990.
Non-participation	N/A, these are archival data.
Randomization	N/A, this is an observational study using existing bibliographic data.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☐ Antibodies
☐	☐ Eukaryotic cell lines
☐	☐ Palaeontology and archaeology
☐	☐ Animals and other organisms
☐	☐ Clinical data
☐	☐ Dual use research of concern
☐	☐ Plants

Methods

n/a	Involved in the study
☐	☐ ChIP-seq
☐	☐ Flow cytometry
☐	☐ MRI-based neuroimaging

Antibodies

Antibodies used — *Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.*

Validation — *Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.*

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s) — *State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.*

Authentication — *Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.*

Mycoplasma contamination — *Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.*

Commonly misidentified lines (See ICLAC register) — *Name any commonly misidentified cell lines used in the study and provide a rationale for their use.*

Palaeontology and Archaeology

Specimen provenance — *Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.*

Specimen deposition — *Indicate where the specimens have been deposited to permit free access by other researchers.*

Dating methods — *If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.*

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight — *Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals — *For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.*

Wild animals — *Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.*

Reporting on sex — *Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall*

	numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.
Field-collected samples	*For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.*
Ethics oversight	*Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	*Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.*
Study protocol	*Note where the full trial protocol can be accessed OR if not available, explain why.*
Data collection	*Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.*
Outcomes	*Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.*

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
☐	☐	Public health
☐	☐	National security
☐	☐	Crops and/or livestock
☐	☐	Ecosystems
☐	☐	Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes	
☐	☐	Demonstrate how to render a vaccine ineffective
☐	☐	Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐	Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐	Increase transmissibility of a pathogen
☐	☐	Alter the host range of a pathogen
☐	☐	Enable evasion of diagnostic/detection modalities
☐	☐	Enable the weaponization of a biological agent or toxin
☐	☐	Any other potentially harmful combination of experiments and agents

Plants

Seed stocks — *Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*

Novel plant genotypes — *Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*

Authentication — *Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

ChIP-seq

Data deposition

☐ Confirm that both raw and final processed data have been deposited in a public database such as GEO.

☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links
May remain private before publication.
For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.

Files in database submission — *Provide a list of all files available in the database submission.*

Genome browser session
(e.g. UCSC)
Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.

Methodology

Replicates — *Describe the experimental replicates, specifying number, type and replicate agreement.*

Sequencing depth — *Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.*

Antibodies — *Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.*

Peak calling parameters — *Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.*

Data quality — *Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.*

Software — *Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.*

Flow Cytometry

Plots

Confirm that:

☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☐ All plots are contour plots with outliers or pseudocolor plots.

☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation — *Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.*

Instrument — *Identify the instrument used for data collection, specifying make and model number.*

Software — *Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.*

Cell population abundance

Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.

Gating strategy

Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type

Indicate task or resting state; event-related or block design.

Design specifications

Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.

Behavioral performance measures

State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

Imaging type(s)

Specify: functional, structural, diffusion, perfusion.

Field strength

Specify in Tesla

Sequence & imaging parameters

Specify the pulse sequence type (gradient echo, spin echo, etc.), imaging type (EPI, spiral, etc.), field of view, matrix size, slice thickness, orientation and TE/TR/flip angle.

Area of acquisition

State whether a whole brain scan was used OR define the area of acquisition, describing how the region was determined.

Diffusion MRI ☐ Used ☐ Not used

Preprocessing

Preprocessing software

Provide detail on software version and revision number and on specific parameters (model/functions, brain extraction, segmentation, smoothing kernel size, etc.).

Normalization

If data were normalized/standardized, describe the approach(es): specify linear or non-linear and define image types used for transformation OR indicate that data were not normalized and explain rationale for lack of normalization.

Normalization template

Describe the template used for normalization/transformation, specifying subject space or group standardized space (e.g. original Talairach, MNI305, ICBM152) OR indicate that the data were not normalized.

Noise and artifact removal

Describe your procedure(s) for artifact and structured noise removal, specifying motion parameters, tissue signals and physiological signals (heart rate, respiration).

Volume censoring

Define your software and/or method and criteria for volume censoring, and state the extent of such censoring.

Statistical modeling & inference

Model type and settings

Specify type (mass univariate, multivariate, RSA, predictive, etc.) and describe essential details of the model at the first and second levels (e.g. fixed, random or mixed effects; drift or auto-correlation).

Effect(s) tested

Define precise effect in terms of the task or stimulus conditions instead of psychological concepts and indicate whether ANOVA or factorial designs were used.

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference

Specify voxel-wise or cluster-wise and report all relevant parameters for cluster-wise methods.

(See Eklund et al. 2016)

Correction

Describe the type of correction and how it is obtained for multiple comparisons (e.g. FWE, FDR, permutation or Monte Carlo).

Models & analysis

n/a	Involved in the study
☐	☐ Functional and/or effective connectivity
☐	☐ Graph analysis
☐	☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity	*Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).*
Graph analysis	*Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).*
Multivariate modeling and predictive analysis	*Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.*